M. Hassan Saeed Hashmi

COMPUTER SCIENCE UNDERGRADUATE · LLM APPLICATIONS & FULL-STACK DEVELOPMENT

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EXPERIENCE

Software Engineering Intern — Rahnor Labs

Remote · May–July 2026

- Contributed to AI-powered CRM systems for businesses and SaaS products alongside senior engineers.
- Used structured specifications, repository context, and AI coding tools to support implementation and debugging.

Freelance Web Developer — Independent

2025–Present

- Delivered paid client web projects from requirements and implementation through deployment and handoff using Next.js and MongoDB.
- Managed domain configuration, email integrations, and client documentation; delivered The Global Apparel website with Next.js, React, and Vercel.

SELECTED PROJECTS

WhyTheDiff — Repository Investigation Application

Python · LangGraph · Qdrant · FastAPI · React · Docker

- Built an agentic retrieval application that investigates public GitHub code changes and produces source-linked explanations with documented, inferred, and unresolved findings.
- Implemented GitHub OAuth, account-isolated histories, a bounded job queue, and checkpoint-based recovery; compared one-pass and agent workflows on three labelled Flask cases and documented citation outcomes, model calls, and latency.

Eight-Thousanders RAG Assistant

Python · PostgreSQL · pgvector · FastAPI · React

- Implemented dense and keyword retrieval with reciprocal rank fusion, cross-encoder reranking, and cited answers over a corpus covering 14 mountains.
- Added a retrieval evaluation harness over 20 labelled questions, comparing dense, hybrid, and reranked configurations using recall@5, MRR, and nDCG; built a streaming interface with inspectable sources.

Research Briefing Agent

Python · Gemini API · SQLite · HTTPX

- Built a tool-calling loop for web research and cited Markdown briefings, with validated tool dispatch, execution budgets, and per-stage traces.
- Added SQLite checkpoints and session memory; demonstrated resuming a forcibly terminated run and recalling evidence from earlier sessions.

TECHNICAL SKILLS

Languages	Python, JavaScript, C++
LLM applications	LangChain, LangGraph, retrieval-augmented generation, hybrid retrieval, reranking, tool calling, evaluation, Groq, Gemini
Backend & data	FastAPI, Node.js, PostgreSQL/pgvector, Qdrant, MongoDB, SQLite, REST APIs
Web & tools	React, Next.js, Tailwind CSS, Docker, Git, GitHub, Vercel

EDUCATION

B.Sc. Computer Science — FAST NUCES, Lahore

2024 — Expected June 2028